CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

# Chapter 11: Areas Related to Circles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

## 1 Sector and segment of a circle

**1.** A sector of a circle of radius 21 cm has a central angle of $60°$. Find the length of its arc. (a) 22 cm (b) 44 cm (c) 88 cm (d) 11 cm CBSE 2026, 1 mark

SOLUTION

Arc length $= \frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{6} \times 132 = 22$ cm

**Answer: (a)**

**2.** The hour hand of a clock is 7 cm long. Through what angle does it turn between 7:00 a.m. and 8:10 a.m.? (a) $\left(\frac{35}{4}\right)^\circ$ (b) $\left(\frac{35}{2}\right)^\circ$ (c) $35°$ (d) $70°$ CBSE 2026, 1 mark

SOLUTION

The hour hand turns $360°$ in 12 hours, which is $30°$ in an hour or $\frac{1}{2}^\circ$ in a minute.

From 7:00 to 8:10 is 70 minutes: $70 \times \frac{1}{2} = 35°$. (The length of the hand is not needed.)

**Answer: (c)**

**3.** A circular ornament of diameter 35 cm is made from silver wire. Wire is also used for 5 diameters, which cut the circle into 10 equal sectors. (i) Find the radius. (ii) Find the circumference. (iii) Find the total length of wire needed, OR the area of each sector. CBSE 2026, 4 marks

SOLUTION

(i) Radius $= \frac{35}{2}$. **Radius** $= 17.5$ **cm.**

(ii) Circumference $= 2 \times \frac{22}{7} \times \frac{35}{2}$. **Circumference** $= 110$ **cm.**

(iii) Wire $=$ circumference $+ 5$ diameters $= 110 + 5 \times 35$. **Total wire** $= 285$ **cm.**

OR: area of the circle $= \frac{22}{7} \times \frac{35}{2} \times \frac{35}{2} = 962.5$ cm², and each sector is one-tenth of it. **Area of each sector** $= 96.25$ **cm².**

**4.** A sector with a central angle of $72^\circ$ has an area of $40\pi$ square units. Find the radius of the circle. CBSE 2025, 1 mark

SOLUTION

$\frac{72}{360} \times \pi r^2 = 40\pi$, so $\frac{r^2}{5} = 40$

$r^2 = 200$

**Answer:** $r = 10\sqrt{2}$ **units**

**5.** A wire 20 cm long is bent into an arc of a circle of radius $\frac{60}{\pi}$ cm. What angle does the arc subtend at the centre? CBSE 2025, 1 mark

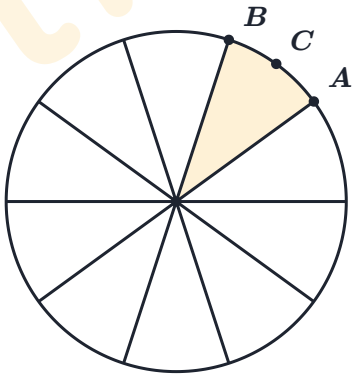
SOLUTION

Arc length $= \frac{\theta}{360} \times 2\pi r$, so $20 = \frac{\theta}{360} \times 2\pi \times \frac{60}{\pi} = \frac{\theta}{360} \times 120$

$\theta = \frac{20 \times 360}{120}$

**Answer:** $60^\circ$

**6.** A circular ornament of diameter 35 mm is made from silver wire. Wire also forms 5 diameters, which cut the circle into 10 equal sectors. $A$ and $B$ are the ends of one sector on the circle and $C$ lies on the arc between them. Take $\pi = \frac{22}{7}$. (i) Find the central angle of each sector. (ii) Find the length of the arc $ACB$. (iii) Find the area of each sector, OR the total length of wire used. CBSE 2025, 4 marks




SOLUTION

(i) Ten equal sectors share $360^\circ$. **Each central angle is** $36^\circ$.

(ii) Radius $r = \frac{35}{2}$ mm, so the circumference is $2 \times \frac{22}{7} \times \frac{35}{2} = 110$ mm. Arc $ACB = \frac{36}{360} \times 110$. **Arc** $ACB = 11$ **mm.**

(iii) Area of the circle $= \frac{22}{7} \times \frac{35}{2} \times \frac{35}{2} = 962.5$ mm². Each sector is one-tenth of it. **Area of each sector** $= 96.25$ **mm².**

OR: wire used = circumference + 5 diameters $= 110 + 5 \times 35$. **Total wire** $= 285$ **mm.**

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**7.** In a circle of radius 21 cm, an arc subtends $60^\circ$ at the centre. Find (i) the length of the arc (ii) the area of the minor segment cut off by the chord of this arc. CBSE 2024, 5 marks

SOLUTION

(i) Length of arc $= \frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{6} \times 132 = 22$ cm

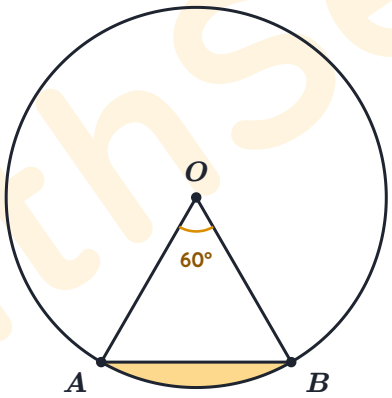
(ii) Area of sector $= \frac{60}{360} \times \frac{22}{7} \times 21 \times 21 = 231$ cm²

The triangle formed by the two radii and the chord has an angle of $60^\circ$ between equal sides, so it is equilateral with side 21 cm. Its area $= \frac{\sqrt{3}}{4} \times 21^2 = \frac{441\sqrt{3}}{4}$ cm²

Area of minor segment $= \left(231 - \frac{441\sqrt{3}}{4}\right)$ cm² $\approx 231 - 190.95 = 40.05$ cm² (taking $\sqrt{3} = 1.732$)

**Answer: (i) 22 cm (ii)** $\left(231 - \frac{441\sqrt{3}}{4}\right)$ **cm²** $\approx 40.05$ **cm²**

**8.** A chord of a circle of radius 14 cm subtends an angle of $60^\circ$ at the centre. Find the area of the minor segment and the area of the major segment. Take $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.732$. CBSE 2023, 5 marks




SOLUTION

Area of the sector $= \frac{60}{360} \times \frac{22}{7} \times 14 \times 14 = \frac{616}{6} \approx 102.67$ cm²

The triangle formed by the two radii and the chord has a $60^\circ$ angle between equal sides, so it is equilateral with side 14 cm. Its area is $\frac{\sqrt{3}}{4} \times 14^2 = 49\sqrt{3} \approx 84.87$ cm².

Minor segment $= 102.67 - 84.87 = 17.80$ cm²

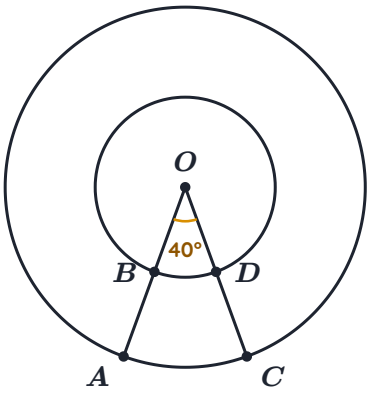
Area of the circle $= \frac{22}{7} \times 14 \times 14 = 616$ cm², so the major segment $= 616 - 17.80 = 598.20$ cm².

**Answer: minor segment** $\approx 17.80$ **cm²; major segment** $\approx 598.20$ **cm²**

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**9.** Two concentric circles with centre $O$ have radii 7 cm and 14 cm. A sector of angle $40^\circ$ is cut away from the ring between them. Find the area of the part of the ring that remains. Take $\pi = \frac{22}{7}$.

CBSE 2016, 3 marks




SOLUTION

The remaining part of the ring corresponds to an angle of $360^\circ - 40^\circ = 320^\circ$.

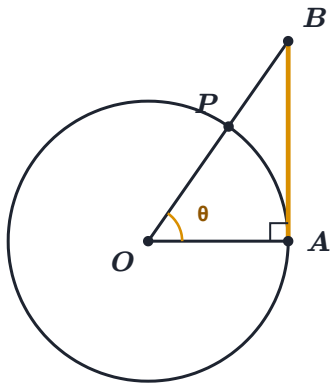
Area $= \frac{320}{360} \times \pi(R^2 - r^2) = \frac{8}{9} \times \frac{22}{7} \times (14^2 - 7^2) = \frac{8}{9} \times \frac{22}{7} \times 147$

$= \frac{8}{9} \times 462 = \frac{3696}{9}$

**Answer: about 410.67 cm²**

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**10.** $OAP$ is a sector of a circle with centre $O$, radius $r$ and angle $\theta$ (in degrees). $AB$ is perpendicular to the radius $OA$ and meets $OP$ produced at $B$. Prove that the perimeter of the region bounded by $AB$, $BP$ and the arc $PA$ is $r\left[\tan\theta + \sec\theta + \frac{\pi\theta}{180} - 1\right]$. CBSE 2016, 4 marks




SOLUTION

In right triangle $OAB$: $\tan\theta = \frac{AB}{r}$, so $AB = r\tan\theta$, and $\sec\theta = \frac{OB}{r}$, so $OB = r\sec\theta$.

$BP = OB - OP = r\sec\theta - r$

Arc $PA = \frac{\theta}{360} \times 2\pi r = \frac{\pi r\theta}{180}$

Perimeter $= AB + BP + \text{arc } PA = r\tan\theta + r\sec\theta - r + \frac{\pi r\theta}{180}$

$= r\left[\tan\theta + \sec\theta + \frac{\pi\theta}{180} - 1\right]$. **Proved.**

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